

LET H_0 BE A COMPRESSED RO WITH DATABASE D ,
INITIALIZED TO EMPTY.

NOW, LET B DO THE FOLLOWING:

1. FOR SOME x^* , MAKE A CLASSICAL QUERY $h \leftarrow H_0(x^*)$.

— THEN, D CONTAINS THE TUPLE (x^*, h) (IN THE PRIMAL DOMAIN)

2. DEFINE $H_b(X, Y)$ AS FOLLOWS:

1. $(X, Y) \leftarrow H_0(X, Y)$ // FORWARD REGISTERS TO H_0

LET $C_x U$ DENOTE THE CONTROLLED
APPLICATION OF U GIVEN x , AND

FOR SOME z , LET:

$$U_z |x, y\rangle = |x, y \oplus z \oplus h\rangle$$

[2. APPLY $C_x U_z$ TO (X, Y) .] // H_1 ONLY

3. APPLY $C_x X$ TO ANCILLA REGISTER **bad**,
INITIALIZED TO $|bad\rangle = |0\rangle$.

THEN, H_0 AND H_1 ARE IDENTICAL EXCEPT ON POINT x^* ,
WHERE $H_0(x^*) = h$ AND $H_1(x^*) = z$.

NOW, LET A PLAY A GAME WHERE THEY ARE SUPPOSED TO
DISTINGUISH THE REAL ORACLE H_0 FROM THE REPROGRAMMED
ORACLE H_1 . ASSUME A HAS AN OPTIMAL STRATEGY, SO
THAT IF THEY SEE z THEY ALWAYS WIN, AND OTHERWISE
THEY MAKE A UNIFORM GUESS.

INSTEAD OF MEASURING A , LET US KEEP THE SUPERPOSITION UNTIL AND AFTER IT HALTS, AND W.L.O.G (I THINK??) ASSUME THAT, WHEN MAKING A UNIFORM GUESS, A SETS ITS OUTPUT REGISTER $|\hat{b}\rangle = |+\rangle = \frac{1}{\sqrt{2}}(|b\rangle + |7b\rangle)$

THE QUANTUM STATE WE END UP WITH IS SOME ENTANGLED STATE

$$|\psi\rangle = \sum_i \alpha_i |\hat{b}\rangle |A_i\rangle |D_i\rangle |bad\rangle.$$

LET US MEASURE THE $|bad\rangle$ REGISTERS, DEFINED VIA PROJECTORS $\{P_{bad}, 1 - P_{bad}\}$ AS USUAL, AND LET $P_{bad}|\psi\rangle = \alpha_{bad}$.

FROM THE SETUP, WE SEE THAT WE HAVE:

$$\begin{aligned} |\psi\rangle &= \alpha_{bad} \left(\sum_i \alpha_i \overset{\hat{b}=b}{|b\rangle} |A_i\rangle |D_i\rangle |1\rangle \right) \\ &\quad + \sqrt{1 - |\alpha_{bad}|^2} \left(\sum_i \alpha_i |+\rangle |A_i\rangle |D_i\rangle |0\rangle \right) \\ &= \alpha_{bad} \left(\sum_i \alpha_i |b\rangle |A_i\rangle |D_i\rangle |1\rangle \right) \\ &\quad + \sqrt{\frac{1 - |\alpha_{bad}|^2}{2}} \left(\sum_i \alpha_i |b\rangle |A_i\rangle |D_i\rangle |0\rangle \right) \\ &\quad + \sqrt{\frac{1 - |\alpha_{bad}|^2}{2}} \left(\sum_i \alpha_i |7b\rangle |A_i\rangle |D_i\rangle |0\rangle \right) \end{aligned}$$

$$\begin{aligned}
&= \alpha_{\text{bad}} \left(\sum_i \alpha_i |b\rangle |A_i\rangle |D_i\rangle |1\rangle \right) \\
&+ \sqrt{\frac{1 - |\alpha_{\text{bad}}|^2}{2}} \left(\sum_i \alpha_i |b\rangle |A_i\rangle |D_i\rangle |0\rangle \right) \\
&+ \sqrt{\frac{1 - |\alpha_{\text{bad}}|^2}{2}} \left(\sum_i \alpha_i |\neg b\rangle |A_i\rangle |D_i\rangle |0\rangle \right).
\end{aligned}$$

READING OFF THE STATE, WE SEE THAT A'S AMPLITUDE FOR WINNING IS

$$\alpha_{\text{win}} := \alpha_{\text{bad}} + \sqrt{\frac{1 - |\alpha_{\text{bad}}|^2}{2}}.$$

$$\begin{aligned}
\Rightarrow |\alpha_{\text{win}}|^2 &= |\alpha_{\text{bad}}|^2 + (\alpha_{\text{bad}}^* + \alpha_{\text{bad}}) \sqrt{\frac{1 - |\alpha_{\text{bad}}|^2}{2}} \\
&\quad + \frac{1 - |\alpha_{\text{bad}}|^2}{2}
\end{aligned}$$

Let $\alpha_{\text{bad}} = |\alpha_{\text{bad}}| e^{i\theta}$,

AND LET $|\psi\rangle \rightarrow e^{i(\frac{\pi}{2} - \theta)} |\psi\rangle$. (Rotate unobservable phase.)

THEN, α_{bad} WILL BE PURELY IMAGINARY, AND $(\alpha_{\text{bad}}^* + \alpha_{\text{bad}}) = 0$.

$$\Rightarrow |\alpha_{\text{win}}|^2 = |\alpha_{\text{bad}}|^2 + \frac{1 - |\alpha_{\text{bad}}|^2}{2}$$

$$\Rightarrow \Pr[\text{bad}] = |\alpha_{\text{bad}}|^2 = 2|\alpha_{\text{win}}|^2 - 1 := \text{Adv}(A).$$



THEN, IF $b_{ad} = \text{TRUE}$, MEASURING D YIELDS A LIST GUARANTEED TO CONTAIN x^* .

IF YOU KNOW WHAT YOU'RE LOOKING FOR, YOU GET TIGHTNESS:

$$\text{Adv}(A) \leq \text{Adv}(B),$$

AND OTHERWISE YOU HAVE TO GUESS, AND

$$\text{Adv}(A) \leq \frac{1}{2} + \text{Adv}(B),$$

MATCHING THE CLASSICAL CASE.

THE IDEA CAN EASILY BE EXTENDED TO SEVERAL x_i^* , AND THE PROGRAMMING POINTS CAN PROBABLY BE MADE ADAPTIVE WITH PERMUTATIONS.

TOO GOOD TO BE TRUE??